개발 환경 구성-v1.4.0

cuda : 12.5

tensorRT : 10.1.0.27

Ubuntu : 22.04

🐳 Docker File 작성

💻 기본 환경 설치

```
#Install nvidia driver
sudo ubuntu-drivers autoinstall

#Install docker
sudo apt-get update
sudo apt-get install curl
curl -fsSL https://get.docker.com/ | sudo sh

sudo chown root:docker /var/run/docker.sock
sudo groupadd docker
sudo usermod -aG docker $USER

distribution=$(. /etc/os-release;echo $ID$VERSION_ID)
curl -s -L https://nvidia.github.io/nvidia-docker/gpgkey | sudo apt-key add -
curl -s -L https://nvidia.github.io/nvidia-docker/$distribution/nvidia\
-docker.list | sudo tee /etc/apt/sources.list.d/nvidia-docker.list
sudo apt-get update && sudo apt-get install -y nvidia-container-toolkit
sudo systemctl restart docker

#(Option)Download tensorrt 8.6.1 for nvidia-L4
https://developer.nvidia.com/nvidia-tensorrt-8x-downloa
```

Docker Build

```
export MS_AI_VERSION=1.4.0
docker build -f ./ms-ai-v1.3.0.dockerfile --no-cache -t \
ms-ai:$MS_AI_VERSION .

export MS_AI_VERSION=1.4.0
export ENV_name=ms-ai
xhost + local:docker
sudo docker run -it \
    --privileged \
    --gpus all \
```

```
    --name $ENV_name \
    --network host \
    --ipc host \
    -v $HOME/workspace:/root/workspace \
    -v $HOME/.Xauthority:/root/.Xauthority \
    -e DISPLAY=$DISPLAY \
    -e QT_X11_NO_MITSHM=1 \
    -v /tmp/.X11-unix:/tmp/.X11-unix:rw \
    -v /dev:/dev \
    -w /root \
    --log-driver=json-file \
      --log-opt max-size=3m \
      --log-opt max-file=5 \
    ms-ai:$MS_AI_VERSION
```

```
export MS_AI_VERSION=1.4.0
docker build -f ./ms-ai-v1.3.0.dockerfile --no-cache -t ms-ai:$MS_AI_VERSION .

export MS_AI_VERSION=1.4.0
export ENV_name=ms-ai-1.4.0
xhost + local:docker
sudo docker run -it \
    --privileged \
    --gpus all \
    --name $ENV_name \
    --network host \
    --ipc host \
    -v $HOME/workspace:/root/workspace \
    -v $HOME/Pictures:/root/Pictures \
    -v $HOME/Videos:/root/Videos \
    -v $HOME/.Xauthority:/root/.Xauthority \
    -e DISPLAY=$DISPLAY \
    -e QT_X11_NO_MITSHM=1 \
    -v /tmp/.X11-unix:/tmp/.X11-unix:rw \
    -v /dev:/dev \
    -w /root \
    --log-driver=json-file \
      --log-opt max-size=3m \
      --log-opt max-file=5 \
    ms-ai:$MS_AI_VERSION
```

```
export ENV_name=test
xhost + local:docker

sudo docker run -it \
    --privileged \
    --gpus all \
    --name $ENV_name \
    --network host \
    --ipc host \
    -v $HOME/workspace:/root/workspace \
    -v /media/$USER/DB_1/:/root/DB_1 \
    -v $HOME/.Xauthority:/root/.Xauthority \
    -e DISPLAY=$DISPLAY \
    -e QT_X11_NO_MITSHM=1 \
    -v /tmp/.X11-unix:/tmp/.X11-unix:rw \
```

```
  -v /dev:/dev \
  -w /root \
  --log-driver=json-file \
    --log-opt max-size=3m \
    --log-opt max-file=5 \
  ms-ai:1.3.0
```

우분투 시작프로그램 설정

```
/etc/systemd/system/docker_python_script.service 파일 생성


[Unit]
Description=Run Python script inside Docker container
After=docker.service
Requires=docker.service

[Service]
ExecStart=/usr/bin/docker start 컨테이너_이름 && /usr/bin/docker \
exec -d 컨테이너_이름 python3 /app/script.py
Restart=always
User=root

[Install]
WantedBy=multi-user.target




서비스 활성화
sudo systemctl daemon-reload
sudo systemctl enable docker_python_script.service
sudo systemctl start docker_python_script.service


___-------
[Unit]
Description=Run Python script inside Docker container
After=docker.service
Requires=docker.service

[Service]
ExecStartPre=/usr/bin/docker start ms-ai
ExecStartPre=/bin/sleep 5
ExecStart=/usr/bin/docker exec ms-ai bash \
-c "cd /root/workspace/MS-AI_1000/v1.4.0-dev/ && python3 main.py"
Restart=always
User=root
TimeoutStartSec=0

[Install]
WantedBy=multi-user.target
```

GUI 실행 필요 라이브러리(우분투)

테스트

```
gst-launch-1.0 rtspsrc location=rtsp://admin:1234@117.17.159.143/normal8 \
latency=200 ! decodebin ! autovideosink
```